
KL Divergence Demystified

KL 发散解密

What does KL stand for? Is it a distance measure? What does it mean to measure the similarity of two probability distributions?
KL 代表什么？它是一种距离度量吗？测量两个概率分布的相似性意味着什么？



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来源： <https://www.freeimages.com/photo/blue-strings-4-1542735>

If you want to intuitively understand what the KL divergence is, you are in the right place, I'll demystify the KL divergence for you.
如果您想直观地了解什么是 KL 发散，那就来对地方了，我将为您揭开 KL 发散的神秘面纱。

As I'm going to explain the KL divergence from the information theory point of view, it is required to know the entropy and the cross-entropy concepts to fully apprehend this article. If you are not familiar with them, you may want to read the following two articles: one for the [entropy](#) and the other for the [cross-entropy](#).
由于我将从信息论的角度来解释 KL 分歧，因此需要了解熵和交叉熵的概念，才能充分理解这篇文章。如果您对它们不熟悉，不妨阅读以下两篇文章：一篇是关于熵的，另一篇是关于交叉熵的。

If you are ready, read on.
如果您准备好了，请继续阅读。

What does KL stand for?

KL 代表什么？

KL in the KL divergence stands for **Kullback-Leibler** which represents the following two people:
KL 发散中的 KL 代表库尔贝克-莱伯勒（Kullback-Leibler），代表以下两个人：



Solomon Kullback (Source: [Wikipedia](#))
所罗门-库尔贝克（来源：维基百科）



Richard A. Leibler (Source: [NSA](#))

理查德-A-莱布勒（资料来源：美国国家安全局）

They introduced the concept of the KL divergence in 1951 ([Wikipedia](#)).

他们于 1951 年提出了 KL 发散的概念（[维基百科](#)）。

What is the KL divergence?

什么是 KL 发散？

The KL divergence tells us how well the probability distribution Q approximates the probability distribution P by calculating the cross-entropy minus the entropy.

KL 发散通过计算交叉熵减去熵，告诉我们概率分布 Q 与概率分布 P 的近似程度。

$$D_{KL}(P||Q) = H(P, Q) - H(P)$$

As a reminder, I put the cross-entropy and the entropy formula as below:

为了提醒大家，我把交叉熵和熵的计算公式写在下面：

$$H(P, Q) = \mathbb{E}_{x \sim P}[-\log Q(x)]$$

$$H(P) = \mathbb{E}_{x \sim P}[-\log P(x)]$$

The KL divergence can also be expressed in the expectation form as follows:

KL 发散也可以用期望形式表示如下：

$$\begin{aligned} D_{KL}(P||Q) &= \mathbb{E}_{x \sim P}[-\log Q(x)] - \mathbb{E}_{x \sim P}[-\log P(x)] \\ &= \mathbb{E}_{x \sim P}[-\log Q(x) - (-\log P(x))] \\ &= \mathbb{E}_{x \sim P}[-\log Q(x) + \log P(x)] \\ &= \mathbb{E}_{x \sim P}[\log P(x) - \log Q(x)] \\ &= \mathbb{E}_{x \sim P}[\log \frac{P(x)}{Q(x)}] \end{aligned}$$

The expectation formula can be expressed in the discrete summation form or in the continuous integration form:

期望公式可以用离散求和形式或连续积分形式表示：

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$
$$D_{KL}(P||Q) = \int P(x) \log \frac{P(x)}{Q(x)} dx$$

So, what does the KL divergence measure? It measures the similarity (or dissimilarity) between two probability distributions.

那么，KL 背离测量的是什么呢？它衡量两个概率分布之间的相似性（或不相似性）。

If so, is the KL divergence a distance measure?

如果是，KL 发散是距离测量吗？

To answer this question, let's see a few more characteristics of the KL divergence.

要回答这个问题，让我们来看看 KL 发散的更多特征。

The KL divergence is non-negative

KL 发散为非负

The KL divergence is non-negative. An intuitive proof is that:
KL 发散为非负。一个直观的证明是

- if $P=Q$, the KL divergence is zero as:
如果 $P=Q$, 则 KL 分歧为零, 因为

$$\log \frac{P}{Q} = \log 1 = 0$$

- if $P \neq Q$, the KL divergence is positive because the entropy is the minimum average lossless encoding size.
如果 $P \neq Q$, 则 KL 发散为正, 因为熵是最小的平均无损编码大小。

So, the KL divergence is a non-negative value that indicates how close two probability distributions are.
因此, KL 分歧是一个非负值, 表示两个概率分布的接近程度。

It does sound like a distance measure, doesn't it? But it is not.
这听起来像是一种距离测量方法, 不是吗? 但事实并非如此。

The KL divergence is asymmetric

KL 发散是不对称的

The KL divergence is not symmetric:
KL 发散不是对称的:

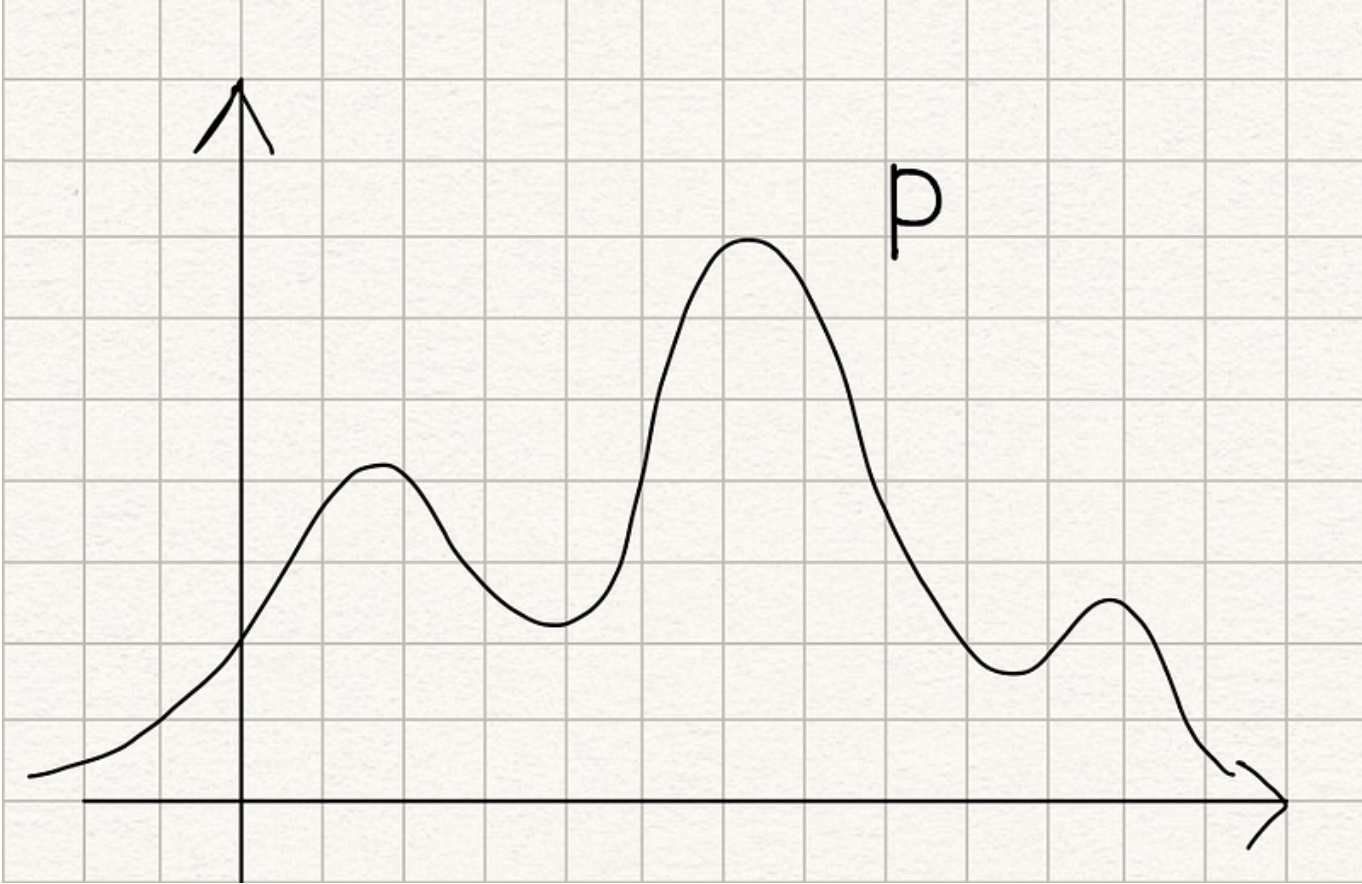
$$D_{KL}(P||Q) \neq D_{KL}(Q||P)$$

It can be deduced from the fact that the cross-entropy itself is asymmetric. The cross-entropy $H(P, Q)$ uses the probability distribution P to calculate the expectation. The cross-entropy $H(Q, P)$ uses the probability distribution Q to calculate the expectation.
这可以从交叉熵本身是不对称的这一事实中推导出来。交叉熵 $H(P, Q)$ 使用概率分布 P 来计算期望值。交叉熵 $H(Q, P)$ 使用概率分布 Q 来计算期望值。

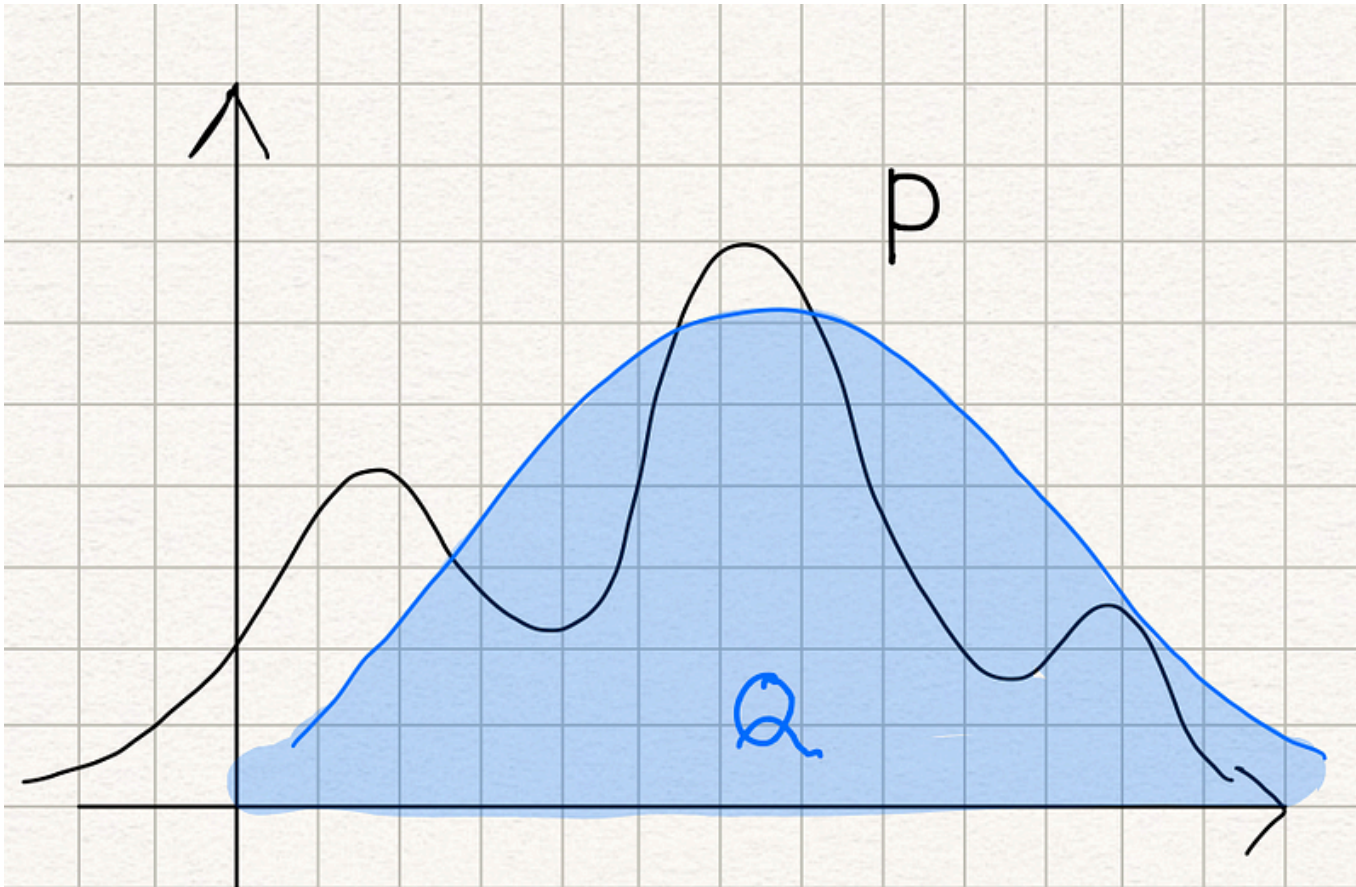
So, the KL divergence cannot be a distance measure as a distance measure should be symmetric.
因此, KL 发散不能作为距离度量, 因为距离度量应该是对称的。

This asymmetric nature of the KL divergence is a crucial aspect. Let's look at two examples to understand it intuitively.
KL 发散的这种非对称性质是一个至关重要的方面。让我们看两个例子来直观地理解它。

Suppose we have a probability distribution P which looks like below:
假设我们有一个如下所示的概率分布 P :



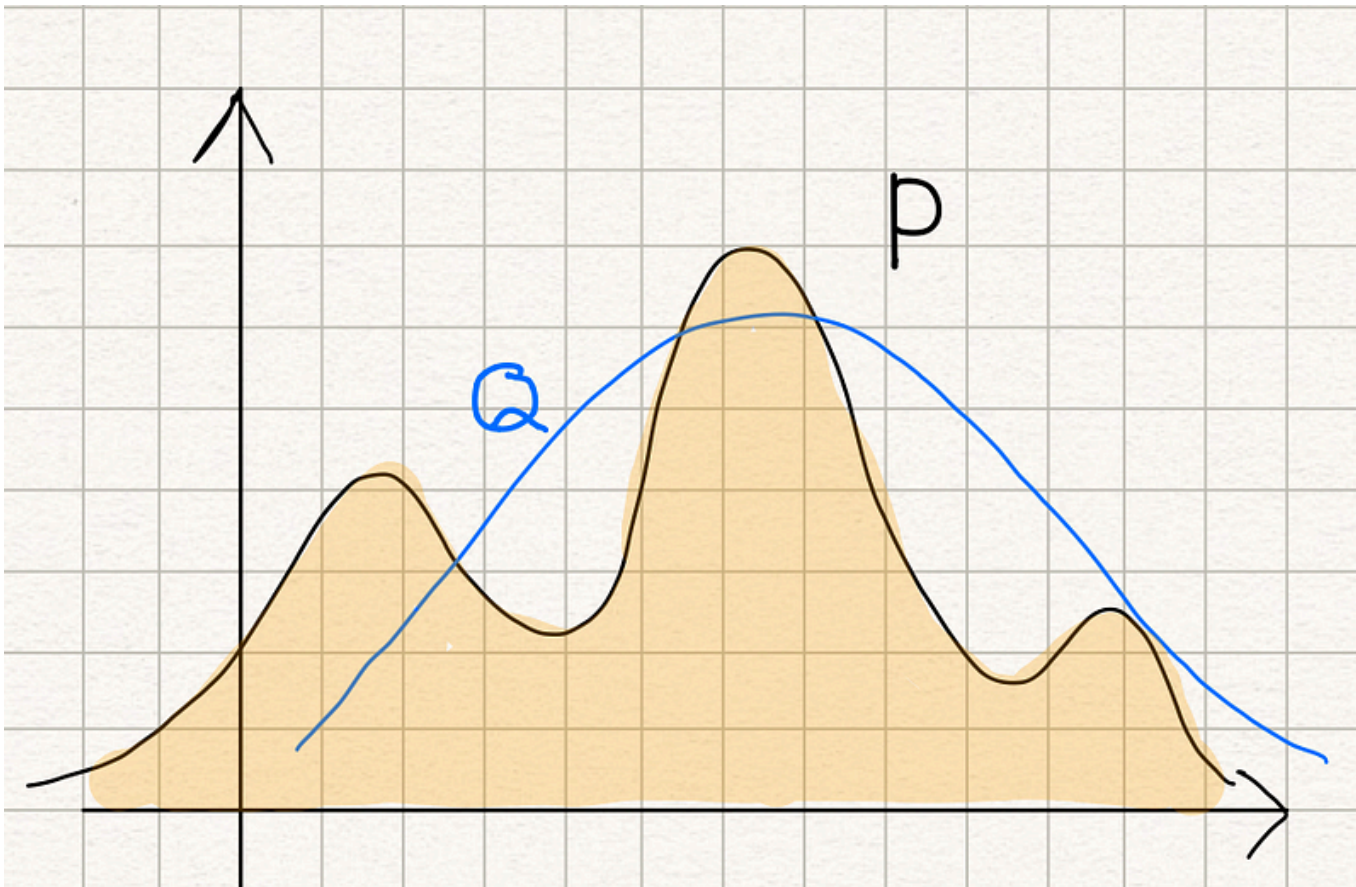
Now, we want to approximate it with a normal distribution Q as below:
现在, 我们想用正态分布 Q 对其进行近似, 如下所示:



The KL divergence is the measure of inefficiency in using the probability distribution Q to approximate the true probability distribution P .
 KL 发散是衡量使用概率分布 Q 近似真实概率分布 P 的效率低下程度的指标。

$$D_{KL}(P||Q) = \mathbb{E}_{x \sim P} \left[\log \frac{P(x)}{Q(x)} \right]$$

If we swap P and Q , it means that we use the probability distribution P to approximate the normal distribution Q , and it'd look like below:
 如果我们把 P 和 Q 互换，就意味着我们用概率分布 P 来近似正态分布 Q ，如下图所示：



$$D_{KL}(Q||P) = \mathbb{E}_{x \sim Q} \left[\log \frac{Q(x)}{P(x)} \right]$$

Both cases measure the similarity between P and Q , but the result could be entirely different, and they are both useful.
 这两种情况都是测量 P 和 Q 之间的相似性，但结果可能完全不同，而且都很有用。

Modeling a true distribution

模拟真实分布

By approximating a probability distribution with a well-known distribution like the normal distribution, binomial distribution, etc., we are modeling the true distribution with a known one.

用正态分布、二项分布等众所周知的分布来逼近概率分布，就是用已知的分布来模拟真实的分布。

This is when we are using the below formula:
这时，我们就要使用下面的公式：

$$D_{KL}(P||Q) = \mathbb{E}_{x \sim P}[\log \frac{P(x)}{Q(x)}]$$

Calculating the KL divergence, we can find the model (the distribution and the parameters) that fits the true distribution well.
通过计算 KL 发散，我们可以找到非常符合真实分布的模型（分布和参数）。

Variational Auto-encoder 变异自动编码器

An example of using the below formula is the variational auto-encoder.
使用下式的一个例子是变分自动编码器。

$$D_{KL}(Q||P) = \mathbb{E}_{x \sim Q}[\log \frac{Q(x)}{P(x)}]$$

I will lightly touch on this topic here as it requires a lot more to explain for people who are not familiar with the variational auto-encoder.
对于不熟悉变分自动编码器的人来说，这需要更多的解释，因此我在这里只简单谈谈这个话题。

The KL divergence is used to force the distribution of latent variables to be a normal distribution so that we can sample latent variables from the normal distribution. As such, the KL divergence is included in the loss function to improve the similarity between the distribution of latent variables and the normal distribution.
KL 发散用于将潜变量的分布强制为正态分布，这样我们就可以从正态分布中对潜变量进行采样。因此，KL 发散包含在损失函数中，以改善潜变量分布与正态分布之间的相似性。

VAE: Variational Auto-Encoder (2013)

Ever stumbled upon the Auto-Encoding Variational Bayes paper and felt lost amid mathematical equations and concepts...

kikaben.com

A few minor mathy points 几个数学上的小问题

The term $p \log p$ becomes zero when p goes to zero.
当 p 为零时， $p \log p$ 项变为零。

$$\lim_{p \rightarrow 0} p \log p = 0$$

When $p > 0$ but $q = 0$, it is defined as infinity.
当 $p > 0$ 但 $q = 0$ 时，它被定义为无穷大。

$$D_{KL}(P||Q) = \infty \quad (where \quad P > 0 \quad but \quad Q = 0)$$

A more rigor proof of the KL divergence being non-negative is as follows:
关于 KL 分歧为非负的更严谨证明如下：

$$\begin{aligned} D_{KL}(P||Q) &= \mathbb{E}_{x \sim P}[\log \frac{P(x)}{Q(x)}] \\ &= \mathbb{E}_{x \sim P}[-\log \frac{Q(x)}{P(x)}] \end{aligned}$$

Since $-\log$ is a convex function, we can apply Jensen's inequality:
由于 $-\log$ 是凸函数，我们可以应用詹森不等式：

$$\begin{aligned}
D_{KL}(P||Q) &\geq -\log(\mathbb{E}_{x\sim P}[\frac{Q(x)}{P(x)}]) \\
&= -\log(\int P(x)\frac{Q(x)}{P(x)}dx) \\
&= -\log(\int Q(x)dx) \\
&= -\log(1) \\
&= 0
\end{aligned}$$

Likelihood Ratio 似然比

There is another way to describe the KL divergence from a probabilistic perspective in that the following likelihood ratio is used.
 还有一种从概率角度描述 KL 发散的方法，即使用下面的似然比。

$$\frac{P(x)}{Q(x)}$$

If you are interested in this approach, I recommend the article by Marko Cotra (the link in the references section below).
 如果您对这种方法感兴趣，我向您推荐马尔科-科特拉（Marko Cotra）的文章（链接见下文参考文献部分）。

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That is all for now. I hope this article is useful to you.
 就写到这里吧。希望这篇文章对你有用。